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Forensic Science 35

Assignment Booklet 4

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Lesson 3	20	
Lesson 4	24	
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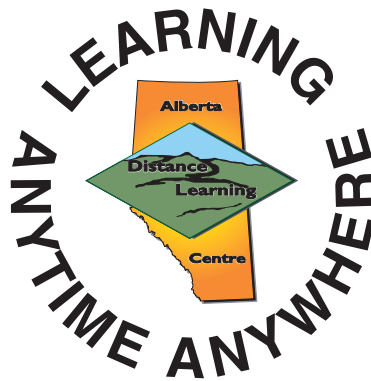
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Forensic Science 35

Assignment Booklet 4

Forensic Ballistics



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Forensic Science 35

Assignment Booklet 4

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For successful completion of this course, you must

- 1. Complete all questions in each Assignment Booklet to the best of your ability.**
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- 2. Achieve at least 40% on the final exam.**
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- 3. Achieve a final course mark of at least 50%.**

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Module 4, Lessons 1 - 4

Total
23

Lesson 1 Assignment: Firearm Basics

Value
11

Part A: Multiple-Choice Questions

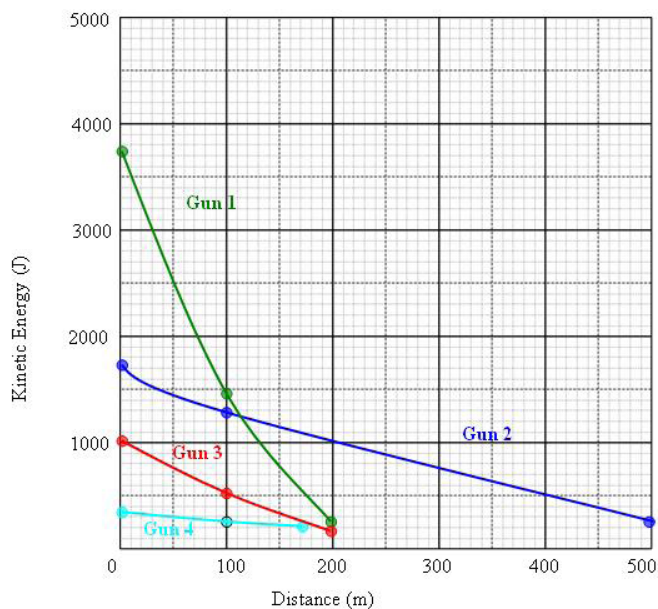
Decide which of the choices best completes the statement or answers the question. Print the letter that corresponds to your choice on the blank next to the question.

- B** 1. Internal ballistics is the study of the evidence produced inside a
- A. cartridge after it has been fired
 - B. firearm when a cartridge is fired
 - C. cartridge as it is fired
 - D. firearm before a cartridge has been found
- C** 2. The two aspects of flight path that are examined during an external ballistics analysis include a bullet's
- A. calibre and target damage
 - B. distance and kinetic energy
 - C. trajectory and maximum range
 - D. calibre and kinetic energy
- C** 3. Which of the following observations is considered part of a terminal ballistics analysis?
- A. Bullet velocity
 - B. Bullet trajectory
 - C. Gunshot wounds
 - D. Firearm discharges
- D** 4. The two main components found within a firearm cartridge are
- A. rim and muzzle
 - B. primer and casing
 - C. trigger and magazine
 - D. bullet and gunpowder

- C** 5. What causes a bullet to be projected from a firearm?
- A. A build-up of smoke within the gun barrel
 - B. Release of smoke from the gun nozzle
 - C. Ignition of gunpowder in the cartridge leading to a controlled explosion
 - D. Release of gunpowder in the cartridge resulting in increased pressure
- C** 6. In comparison to handgun bullets, rifle bullets tend to have ____ muzzle velocities because force is applied to the bullet for a ____ time. The answer that correctly completes this statement is
- A. lower, longer
 - B. lower, shorter
 - C. higher, longer
 - D. higher, shorter
- B** 7. The significant loss of kinetic energy and muzzle velocity in a revolver is due to
- A. decrease in cartridge density after firing
 - B. escape of gases from the cylinder and barrel
 - C. release of gunpowder from the cartridge after ignition
 - D. increase in pressure from the build-up of gases in the muzzle
- D** 8. Which of the following describes one reason modern law enforcement officers use pistols rather than revolvers?
- A. Pistols are more accurate.
 - B. Pistol cartridges have a higher muzzle velocity.
 - C. Pistol cartridges are more accurate due to their low trajectory.
 - D. Pistols hold more cartridges due to a higher magazine capacity.
- B** 9. Which of the following firearms most likely has the best long range target accuracy?
- A. Revolver
 - B. Bolt-action rifle
 - C. Automatic pistol
 - D. 12-gauge shotgun

Use the following graph to answer questions 11 and 12.

The Kinetic Energy of Ammunition from Four Guns



 A 10. The firearm with the most stopping power at a distance of 100m is

- A. gun 1
- B. gun 2
- C. gun 3
- D. gun 4

 B 11. The firearm with the most stopping power at a distance of 150m is

- A. gun 1
- B. gun 2
- C. gun 3
- D. gun 4

Value
3

Part B: Numeric-Response Questions

Use the following information to answer question 1.

Definitions of Various Types of Firearms

1. This gun sprays lead pellets over a large area.
2. Individual cartridges are loaded into a revolving cylinder.
3. Mounted telescopic sights help focus upon a long-range target.
4. Cartridges are held within a magazine inserted into the handle.

①

1. Match each definition above with the correct type of firearm given below.

Description:	<u>4</u>	<u>2</u>	<u>3</u>	<u>1</u>
Firearm:	pistol	revolver	rifle	shotgun

Use the following information to answer question 2.

Capabilities of Various Types of Firearm Bullets

1. Lowest muzzle velocity
2. The greatest amount of kinetic energy when the bullet leaves the gun
3. Best long-range accuracy due to a flatter trajectory

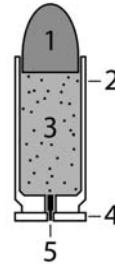
①

2. Match each firearm bullet capability above with the correct type of firearm given below.

Description:	<u>1</u>	<u>3</u>	<u>2</u>
Firearms:	handgun	rifle	shotgun

Use the following diagram to answer question 3.

The Main Parts of a Firearm Cartridge



①

3. Match each part of the firearm cartridge above with the correct name given below.

Part:	<u>5</u>	<u>1</u>	<u>2</u>	<u>3</u>
Name:	primer	bullet	casing	gunpowder

Value
9

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences.

②

1. Describe two differences between a bullet and a cartridge.

• ***A bullet is one of several parts of a cartridge. 1 mark***

• ***When a gun is fired, the bullet is propelled from the barrel and the empty cartridge casing remains behind. 1 mark***

②

2. Discuss why bullets and their casings are often found in different locations at crime scenes.

When a firearm is discharged, the bullet is fired from the gun barrel by the explosive gases released by the ignition of gun powder in the shell casing. Then, the casing is either retained in the firearm or is ejected by the force of the recoil that automatically chambers a new cartridge in the process. Therefore, while the bullet may be lodged in an object at the crime scene, the shell casing may still be in the firearm or may have been ejected, landing somewhere in the immediate area of the shooter's position.

- ③ 3. Using your knowledge of velocity, kinetic energy, and trajectory, explain how a handgun round becomes less effective than a rifle bullet over long distances. *A rifle bullet will leave the rifle with greater kinetic energy and velocity because a larger amount of gunpowder is contained in the cartridge. A handgun round typically suffers from a significant reduction in velocity and kinetic energy as it travels beyond a certain distance. This means a drop in trajectory. A bullet fired from a large calibre rifle retains a large amount of velocity and kinetic energy even after it has travelled one or two kilometres. This enables the bullet to have a flatter trajectory. In contrast, the kinetic energy of a handgun round is expended much more quickly. Consequently, this round will lose height and fall to the ground much sooner.*
- ② 4. Explain why a police tactical unit always takes at least one assault rifle or sniper rifle when responding to a call involving a suspect armed with a gun. *A high-powered rifle is able to fire a bullet over long distances with a high degree of accuracy and kinetic energy. This provides officers with an opportunity to effectively neutralize a target with reduced risk to the public as well as other officers. A scope mounted on the gun allows for more accurate aiming.*

Total
23

Lesson 2 Assignment: Ballistic Fingerprinting and Wound Ballistics

Value
10

Part A: Multiple-Choice Questions

Decide which of the choices best completes the statement or answers the question.
Print the letter that corresponds to your choice on the blank next to the question.

- A 1. The interior of a gun barrel contains numerous unique impressions known as
- A. lands and grooves
 - B. indentations and striations
 - C. extractor and ejector marks
 - D. chamber marks and fingerprints
- C 2. Rifling increases the accuracy of a firearm by
- A. forcing the bullet to rotate as it leaves the barrel
 - B. causing the bullet to expand as it leaves the muzzle
 - C. stabilizing the bullet as it is propelled out of the cylinder
 - D. preventing the bullet from spinning as it is projected from the gun
- D 3. The unique rifling impressions in every gun allow forensic investigators
- A. calculate the exact velocity of a particular bullet
 - B. match a fired bullet to its empty cartridge casing
 - C. determine the likely characteristics of an entry wound
 - D. determine if a bullet has been fired from a particular gun
- C 4. Extractor and ejector marks on cartridge casings are produced
- A. during normal wear and tear
 - B. during the manufacturing process
 - C. as the casing is removed from the chamber
 - D. as the casing is propelled through the gun barrel

- D** 5. The microscope indentations caused by the firing pin when it strikes the primer of a cartridge are called
- A. chamber marks
 - B. primer impressions
 - C. ammunition stamps
 - D. firing pin impressions
- D** 6. Ballistic fingerprints identifies
- A. the type of gunpowder found within a cartridge
 - B. the number of bullets fired by a particular firearm
 - C. the distinct rifling impressions within a gun barrel
 - D. the unique markings on bullets and cartridge casings
- B** 7. As a bullet penetrates a victim's body, the process that causes tissue damage is the
- A. ignition of the primer of the cartridge
 - B. release of kinetic energy from the bullet
 - C. explosion of gunpowder in the gun barrel
 - D. destruction of the casing around the bullet
- C** 8. Which of the following bullet designs expands the most, thus causing severe tissue damage?
- A. Hollow Point
 - B. Wadcutter
 - C. Metal Jacket
 - D. Pointed Soft Point
- A** 9. The type of firearm that causes the most severe wounds in a victim 500m away is a
- A. rifle
 - B. pistol
 - C. shotgun
 - D. revolver

- A 10. Which of the following tissues will likely sustain the least damage when struck by a bullet?

A. Lung
B. Bone
C. Heart
D. Kidney

Value
2

Part B: Numeric-Response Questions

Use the following list to answer question 1.

Descriptions related to Ballistic Fingerprinting

1. Computerized ballistic fingerprinting identification system
2. Parallel striations within a gun barrel
3. Manufacturer information located on cartridge base
4. Parallel striations on a casing

①

1. Match each description above with the correct ballistic fingerprinting term given below.

Description:	<u> 1 </u>	<u> 4 </u>	<u> 2 </u>	<u> 3 </u>
Term:	IBIS	chamber marks	rifling impressions	ammunition stamp

Use the following information to answer question 2.

Descriptions Related to Forensic Ballistics

1. wadcutter
2. hollowpoint
3. handgun
4. military rifle
5. low velocity
6. high velocity

①

2. Use the numbered labels from the list above to complete the following statement:

At a range of 200 metres, the most serious gunshot would result from a 2 bullet fired at 6 from a 4 .

Value
11

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences.

③

1. Identify three unique characteristics that a forensic ballistic expert might analyze while examining a bullet and a cartridge casing to determine the type of gun used at a crime scene.

Bullet: Calibre, wound ballistics 2 marks

Cartridge casing: ammunition stamps 1 mark

②

2. Police officers are trained to shoot at centre mass when faced with lethal threat encounters. Centre mass is considered to be the centre of the chest or the breastbone region. Use your knowledge of wound ballistics to explain why they are trained to aim for this part of the human body.

A hit to centre mass is easiest to achieve when the officer is under stress. It has the greatest chance of incapacitating a suspect who is trying to cause bodily harm to the officer or other individuals. The wound cavity would likely involve massive injury of the heart or other vital organ or cause the rib cage to shatter leading to further injury. Organs are relatively inelastic and are therefore easy to injure. This type of hit may also affect the spinal column.

A hit in the cranial area may be just as effective, but the likelihood of missing the target is great, making this tactic extremely risky when public safety is considered.

Use the following information to answer question 3.

A 20-year old male armed with a handgun shot a rival gang member twice at close range. The victim suffered two large gunshot wounds—one to his left arm and a second wound that severed the large femoral artery in the upper portion of his left leg. Each bullet expanded upon entry and neither bullet exited the victim's body.

- ② 3. a. Identify the type of bullet that likely caused these injuries. Explain your choice.

The bullet used was likely a hollowpoint because such bullets mushroom (expand) upon entry to make a large wound cavity with increased destruction likely.

- ② b. Explain which of the two gunshot wounds most likely caused the victim's death.

The gunshot wound to the upper leg likely caused the victim's death. It served a large artery, which likely caused extensive blood loss.

- ② 4. Discuss how the injuries to President Kennedy and Governor Connally would have been different if Lee Harvey Oswald had used a handgun instead of a rifle.

The injuries would have been less severe to the President and Governor if Oswald had used a handgun. Handgun bullets are propelled at lower velocity and, therefore, have less kinetic energy than rifle bullets.

Total
20

Lesson 3 Assignment: Testing for Gunshot Residue

Value
9

Part A: Multiple-Choice Questions

Decide which of the choices best completes the statement or answers the question.
Print the letter that corresponds to your choice on the blank line next to the question.

- B** 1. The mass use of semi-automatic handguns by law enforcement agencies occurred soon after the development of
- A. ammunition with black powder
 - B. ammunition with smokeless gunpowder
 - C. larger cartridges containing more gunpowder
 - D. smaller cartridges containing less gunpowder
- C** 2. A sample of gunshot residue always consists of
- A. nitrates and nitrites
 - B. charcoal and sulphur
 - C. a propellant and primer
 - D. black powder and smokeless powder
- C** 3. The development of smokeless powder was necessary because black powder
- A. causes a small combustion reaction
 - B. discharges very little high pressure gas
 - C. creates thick smoke and a large amount of soot
 - D. produces a large amount of charcoal and sulphur
- A** 4. The row that correctly identifies the main substances found in black powder and smokeless powder is row

Row	Black Powder	Smokeless Powder
A.	charcoal, sulphur, and potassium nitrate	nitrites
B.	nitrocellulose and nitro-glycerine	lead
C.	barium, lead, and antimony sulphide	nitrates
D.	lead styphnate and potassium chlorate	mercury

- A 5. Smokeless powder produces gas but no smoke because it contains
- A. nitrites
 - B. potassium
 - C. sulphur
 - D. charcoal
- C 6. Which of the following substances may be found in the primer of a cartridge?
- A. Nitrates and nitrites
 - B. Nitrocellulose and nitro-glycerine
 - C. Antimony, barium, lead, and mercury
 - D. Charcoal, potassium, sodium, and sulphur
- A 7. What substance in smokeless powder does the Paraffin test detect?
- A. Nitrates and nitrites
 - B. Nitrocellulose and nitro-glycerine
 - C. Antimony, barium, lead, and mercury
 - D. Charcoal, potassium, sodium, and sulphur

Use the following information to answer questions 8 and 9.

The GSR testing solution was swabbed onto the hands of an armed robbery suspect. The solution turned a brownish-pink in colour.

- D 8. Which of the following GSR tests was most likely used on this suspect?
- A. Paraffin
 - B. Harrison-Gilroy
 - C. Modified Greiss
 - D. Sodium Rhodizonate
- B 9. Which of the following elements was detected by this GSR test?
- A. Lead
 - B. Barium
 - C. Mercury
 - D. Antimony

Value
2

Part B: Numeric-Response Questions

Use the following list to answer question 1.

Gunshot Residue Positive Test Results

1. Blue flecks
2. Red-orange specks
3. Orange colour change
4. Brown-pink colour change

①

1. Match each description above with the correct test given below.

Description:	<u>1</u>	<u>4</u>	<u>3</u>	<u>2</u>
Term:	Paraffin	Sodium Rhodizonate	Harrison- Gilroy	Modified Greiss

Use the following list to answer question 2.

Descriptions of Two Substances and Two Gunshot Residue Tests

1. All modern ammunition contains this substance.
2. This GSR test detects nitrites on surfaces or clothing.
3. During combustion, 45% of its product is a thick smoke.
4. This test can detect and identify any type of element or compound.

①

1. Match each description above with the correct substance or test given below.

Description:	<u>3</u>	<u>1</u>	<u>2</u>	<u>4</u>
Term:	Black Powder	Smokeless Powder	Modified Greiss	X-ray Analysis

Value
9

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences. Use the information below to answer questions 1 and 2.

A piece of gauze turns a scarlet red colour after being swabbed over the hands of a murder suspect.

①

1. a. Before it was swabbed over the suspect's hands, what substance was the gauze dipped into?

a dilute hydrochloric acid

①

- b. What specific element was detected in this test?

lead

②

- c. Identify whether the element detected in this test was from a propellant or a primer. Justify your choice.

primer - Lead is found only in primer

②

2. Why is the scanning electron microscope being used more frequently in GSR testing?

Any of the following for one mark each.

- ***The scanning electron microscope can detect and confirm any type of element, ion, or compound from gunshot residue that may be found on a suspect.***
- ***most reliable method*** • ***low rates of false positive results***
- ***can analyze residues up to 12 hrs after crime***

Both the Paraffin test and Harrison-Gilroy test are performed on an armed robbery suspect. The paraffin test is positive for nitrites, and the Harrison-Gilroy test is positive for antimony. X-ray analysis with a scanning electron microscope confirms the results of both tests.

③

3. Can the combined results of these three tests ensure a conviction in court?

Not necessarily - positive GSR results never alone prove the guilt of a suspect. The paraffin test detects nitrites, which is in smokeless powder, and the Harrison-Gilroy test detects antimony, which is in the primer of cartridge. These results, plus the fact that they have been confirmed by the scanning electron microscope, make more plausible to a jury the conclusion that the suspect had recently used a firearm. It is possible that the suspect merely touched a recently used gun or been contaminated from another source. Must have other valid pieces of evidence to support this.

Total
24

Lesson 4 Assignment: Crime Case Studies Involving Forensic Ballistics

Value
10

Part A: Multiple-Choice Questions

Decide which of the choices best completes the statement or answers the question. Print the letter that corresponds to your choice on the blank line next to the question.

Case Study #1: The Beltway Snipers

- C** 1. Which of the following firearms was used in the Beltway sniper killings?
- A. Bolt-action rifle
 - B. 12-gauge shotgun
 - C. Semi-automatic rifle
 - D. Semi-automatic pistol
- D** 2. The Beltway sniper suspects were able to conceal themselves by firing at their victims from a
- A. car disguised as a police car
 - B. location that was heavily treed
 - C. distance greater than 200 metres
 - D. firing port in the trunk of their car
- D** 3. To compare with evidence found at the Beltway sniper shooting crime scenes, test bullets were fired from the suspect's rifle to produce
- A. bullets and cartridges
 - B. DNA and fingerprints
 - C. gunpowder and primer residue
 - D. bullet fragments and cartridge casings

Use the following information to answer questions 4 and 5.

A forensic chemist found traces of nitro-glycerine in the trunk of John Allen Muhammad's car.

- B** 4. What type of gunshot residue testing technique could have been used to identify the nitro-glycerine found in the suspect's car?
- A. Harrison-Gilroy Test
 - B. Modified Greiss Test
 - C. Potassium Chlorate Test
 - D. Sodium Rhodizonate Test
- D** 5. In what part of a cartridge is nitro-glycerine found?
- A. Bullet
 - B. Primer
 - C. Casing
 - D. Gunpowder

Case Study #2: Distance, Shielding, and Movement Critical Incident Response

- D** 6. When confronting an armed suspect, upon which three factors is it most important for a police officer to focus?
- A. Concealment, lighting conditions, and containment of suspect(s)
 - B. Distance, type of firearm, and the physical characteristics of the suspect(s)
 - C. Public safety, type of suspect ammunition, and preservation of evidence at the crime scene
 - D. Shielding, maintaining a safe distance, and moving to an area from which the suspect can be contained
- B** 7. If a police officer is armed only with a handgun, what is the approximate maximum distance at which he or she could effectively engage a suspect?
- A. 5 m
 - B. 50 m
 - C. 250 m
 - D. 500 m

- A 8. If a police officer is armed only with a military assault rifle, what is the approximate maximum distance at which he or she could effectively engage a suspect?
- A. 100 m
B. 500 m
C. 5 000 m
D. 25 000 m
- C 9. Which of the following objects most likely provides the best shielding for a police officer if faced with a lethal threat from the robbery suspect hiding in the playground?
- A. Park bench
B. Playground slide
C. Car engine block
D. Chain-link fence
- C 10. Which of the following objects most likely provides the best shielding for a police officer if facing a lethal threat from a robbery suspect armed with a shotgun loaded with slugs?
- A. Car door
B. Several small trees
C. 1 m high brick wall
D. 3 m high wooden fence

Value
2

Part B: Numeric-Response Questions

Case Study #1: The Beltway Snipers

Use the following information to answer question 1.

Various Events in the Beltway Sniper Investigations

1. A description of the suspect's car is released to the media.
2. A tree stump from a home in Tacoma, Washington is seized.
3. Lee Boyd Malvo's fingerprint is found at the scene of one of the shootings.
4. Tarot cards and handwritten letters are found at the scene of several shootings.

①

1. The correct sequence of events that occurred during the Beltway sniper investigations is 4, 3, 1, and 2.

Case Study #2: Distance, Shielding, and Movement Critical Incident Response

Use the following information to answer question 2.

Various Poisoning Symptoms

1. Fingerprints found in suspect's car
2. Type of gun obtained from a suspect
3. Rifling impressions on bullets found in victim
4. Gunshot residue found upon a suspect's hands
5. Firing pin impressions on cartridge near victim
6. DNA collected from mask that matches the suspect's DNA

①

1. Evidence that would prove most useful in helping to prove which one of the three suspects killed the convenience store clerk would be 2, 3, 4, and 5.

(Record your answers in ascending order.)

Value
12

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences.

Case Study #1: The Beltway Snipers

③

1. Explain why no intact bullets were found at any of the Beltway sniper shootings.

• *No intact bullets were found in the victims because high velocity bullets explode into bullet fragments.*

• *High-velocity bullets explode into fragments when they hit bone and/or other issue.*

Military snipers are trained to shoot at and hit targets from great distances, typically 500 to 1000 metres.

③

2. Considering the range at which most of the shootings took place, why have some military experts criticized the media's use of the label *Beltway sniper*?

• ***The Beltway sniper's victims were shot at distances of 100 metres or less.***

• ***The name Beltway sniper was likely criticized because the shots were taken from distances far less than those for which military snipers are trained. Hitting a target at less than 500 metres is not difficult with an assault rifle.***

Case Study #2: Distance, Shielding, and Movement Critical Incident Response

③

3. A security guard arrests a man at a shopping mall after he is caught trying to steal clothing. When police arrive, they suspect he is one of the three suspects from the convenience store robbery and murder. How could GSR testing help to confirm their suspicions?

• ***Gunshot residue (GSR) testing could be completed on the suspect to determine if he has fired a gun recently.***

• ***GSR testing would identify any of the unique particles from gunpowder or primer that are released when a gun is fired.***

③

4. Describe three reasons police officers confronting an armed suspect should know the type of firearm the suspect is carrying.

• *Helps them to identify the type of shielding (cover) they need.*

• *Helps to determine a safe distance at which the suspect can be engaged.*

• *Helps to determine the size of the area that needs to be cordoned off to protect the public.*

END OF ASSIGNMENT BOOKLET 4



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